G249 Anlysis Report

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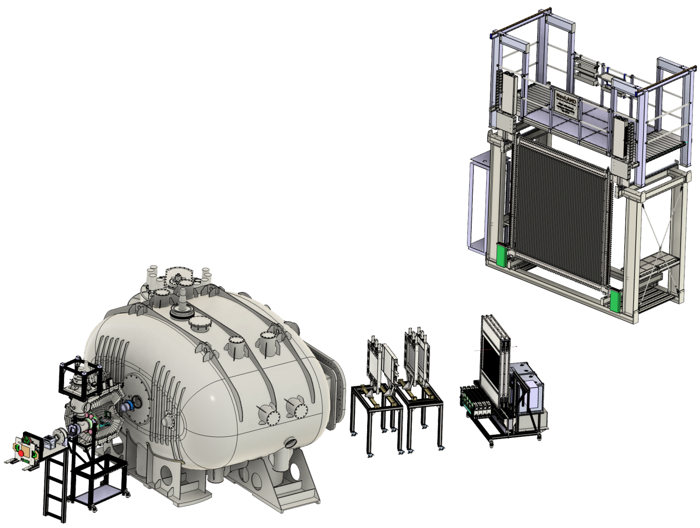

This document is a brief description of the different task performed in the analysis of the G249 experiment. In particular we focus on the work done by the macros of the GitHub repository g249_analysis.

1. Calibrations

These macros check the performance of the calibrations of some detectors (FOOTs and NeuLAND for the moment).

1.1. FOOT

For FOOTs, all the calibrations were done in-flight by R3BROOT (in map2cal and cal2hit classes), so the only correction done after the experiment was charge calibration. The problem is that there is a dependence on the energy with the eta parameter that does not allow to perform a direct conversion between energy and charge (see fig. 1 left).

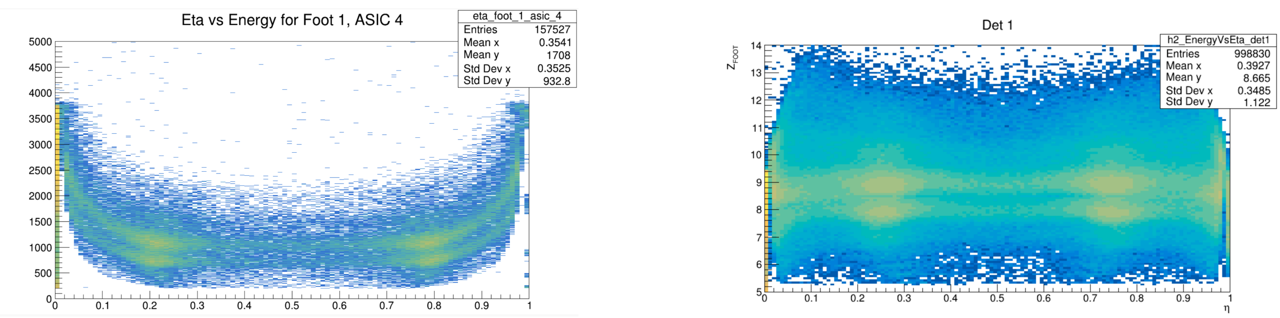

Figura 1: Energy as a function of the eta parameter. Left: With eta correction or charge calibration, only one ASIC. Right: After eta correction and charge calibration for all the ASICs of the detector.

To make the eta correction we perform a high-degree polynomial fit to the $Z = 8$ and $Z = 9$ bands for each ASIC chip and calibrate the energy by making a linear fit (channels, charge) for those bands. This work is done by R3BROOT and, in the macro `check_eta_correction.C`, we plot the charge as a function of eta to show that the former dependence has disappear (see fig. 1 right).

On the other hand, this macro also represent the charge measured in the FOOTs (after all the corrections) against the charge measured in LOS. In fig. 2, we see a clear correlation between both charges. Whereas for charges different from $Z = 8$ and $Z = 9$ we see some saturation, this is not present for our charges of interest.

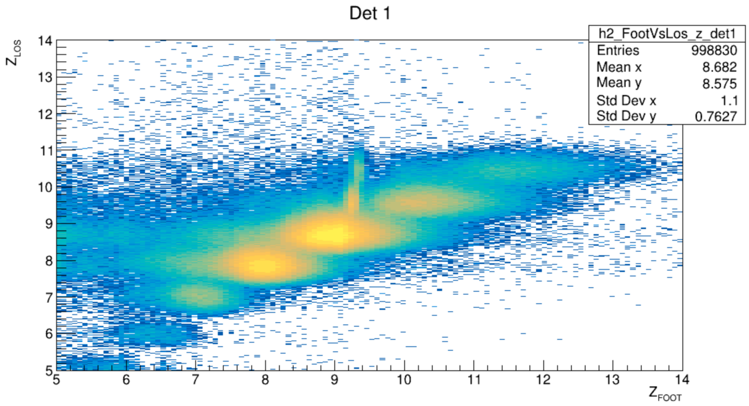

Figura 2: Charge in FOOT vs charge in LOS for the same events

1.2. NeuLAND